

University Social Media & Admissions Analysis

How do TikTok, Instagram & YouTube presence correlate with university admissions outcomes?

Pepperdine University · MSBA Team Project · Spring 2025

51 Schools Surveyed · 38 Fully Merged · IPEDS 2024 + 3 Social Media APIs

The Business Question

Why social media presence matters for universities

"Does a university's social media presence and engagement correlate with admissions outcomes like yield rate, acceptance rate, and application volume?"



Recruit Gen Z

Prospective students discover schools on TikTok and Instagram before visiting a website



Measurable ROI

Engagement rate, follower growth, and views are quantifiable — unlike word of mouth



Brand Positioning

Social content shapes perceived selectivity and culture — influencing enrollment decisions



Competitive Intel

Benchmarking against peer institutions reveals strategic gaps and opportunities in real time

Data & Methodology

Three social media APIs + federal IPEDS data, merged on UNITID school ID

Data Sources

Instagram

RapidAPI (instagram120)

Last 12 posts — followers, avg likes, comments, engagement rate

TikTok

RapidAPI (tiktok-scraper7)

Last 30 posts — followers, avg views, likes, shares, engagement rate

YouTube

Google YouTube Data API v3

Subscribers, total views, view velocity, Shorts %, posting frequency

IPEDS 2024

Federal Education Database

Acceptance rate, yield rate, application volume, SAT/ACT scores

Scope & Key Decisions

► 51 Universities Surveyed

CA private & public + national elite schools

► 38 Fully Merged

Complete data across all 4 sources

► ~30 Posts (TikTok)

Recent posts = most current signal available

► 12 Posts (Instagram)

Reliable avg engagement from most recent content

► UNITID as Join Key

Federal school ID merges all datasets cleanly

► Engagement Rate = Core KPI

$(\text{Likes} + \text{Comments} + \text{Shares}) \div \text{Followers} \times 100$

► Correlation Method

Pearson r · ** $p < 0.05$ · * $p < 0.10$

Instagram: Follower Scale vs. Engagement Quality

High follower counts don't guarantee engagement — Pepperdine proves this

	instagram_handle_x	ig_followers	ig_avg_likes	ig_avg_comments	ig_avg_engagement	ig_engagement_rate
0	pepperdine	64828	1894.3	33.1	1927.4	2.9731
1	uscud	266305	6733.2	130.4	6863.6	2.5773
2	stanford	1380737	2678.9	46.7	2725.6	0.1974
3	santaclarauniversity	45084	1408.7	22.1	1430.8	3.1736
4	loyolamarymount	50432	922.3	17.4	939.8	1.8635
5	caltechedu	89701	428.4	7.0	435.4	0.4854
6	universityofsandiego	64	NaN	NaN	NaN	NaN
7	usfca	49978	728.3	5.6	733.9	1.4684
8	claremontmckennacollege	14947	142.9	1.3	144.2	0.9647
9	pomonacollege	21750	451.5	4.1	455.6	2.0947
10	occidentalcollege	18135	617.7	11.0	628.7	3.4668
11	chapmanu	55816	882.0	20.5	902.5	1.6169
12	whittiercollege	10917	307.3	5.8	313.2	2.8689
13	azusapacific	23109	195.0	9.0	204.0	0.8828
14	biolauniversity	35765	1728.4	34.2	1762.6	4.9283
15	bostonu	243219	2353.8	29.3	2383.2	0.9799
16	scrippscollege	12478	253.9	3.0	256.9	2.0588
17	callutheran	13864	315.1	4.0	319.1	2.3016
18	universityofredlands	18279	312.8	5.2	318.0	1.7397
19	ucberkeleyofficial	416505	13652.5	109.5	13762.0	3.3042
20	ucla	509295	3058.8	68.2	3126.9	0.6140
21	ucsandiego	153583	6794.9	67.5	6862.4	4.4682
22	ucdavis	135419	2650.2	22.8	2673.1	1.9739
23	ucsantabarbara	105682	3292.8	34.7	3327.5	3.1486
24	ucirvine	123226	2613.3	30.2	2643.6	2.1453
25	ucsc	72354	1870.2	13.8	1884.1	2.6040
26	ucriversideofficial	68055	2090.8	37.8	2128.7	3.1279
27	calpoly	97661	1883.2	30.9	1914.2	1.9600
28	sandiegostateuniversity	120184	2515.1	15.2	2530.3	2.1054
29	sjsu	75075	727.0	1.6	728.6	0.9705



Follower Gap

Pepperdine: 64,828 followers

Dataset avg: 286,709

Percentile: 24th

Harvard: 3,039,388 · Stanford: 1,380,737

Yale: 871,039 · UCLA: 509,295



Engagement Strength

Pepperdine: 2.97% eng. rate

Percentile: 87th

Top performers:

Biola: 4.93% · UC San Diego: 4.47%

Occidental: 3.47% · Pepperdine: 2.97%

Harvard: 0.22% · Stanford: 0.20%

TikTok: Massive Scale Variance — Engagement Tells a Different Story

46 of 51 schools active · Followers range from 1,156 to 854,518

	school_name	tt_followers	tt_avg_views	tt_engagement_rate	tt_videos_analyzed
0	Pepperdine University	7943.0	6740.7	4.5544	30
1	University of Southern California	37887.0	11874.1	7.8903	29
2	Stanford University	1156.0	4303.2	7.2713	14
3	Santa Clara University	2392.0	2070.4	2.2170	30
4	Loyola Marymount University	6660.0	3622.3	5.2204	30
5	University of San Diego	5697.0	43878.2	11.1641	30
6	University of San Francisco	1566.0	26622.4	14.9393	30
7	Claremont McKenna College	1293.0	2494.2	3.3959	32
8	Pomona College	2156.0	1728.4	5.4617	29
9	Occidental College	1734.0	1556.6	5.5827	30
10	Chapman University	5997.0	23500.8	2.0842	33
11	Azusa Pacific University	4085.0	360791.7	0.0764	30
12	Biola University	2118.0	2696.8	7.6313	31
13	Boston University	45932.0	16036.1	6.1973	30
14	California Lutheran University	3152.0	117959.1	0.2676	30
15	UC Berkeley	32488.0	253810.4	9.3201	30
16	UCLA	213578.0	464341.7	10.8315	30
17	UC San Diego	23333.0	50826.4	11.8736	30
18	UC Davis	15092.0	14763.4	9.3136	32
19	UC Santa Barbara	19961.0	11872.4	12.1391	30
20	UC Irvine	1226.0	NaN	NaN	0
21	UC Santa Cruz	13841.0	51347.2	16.7789	30
22	UC Riverside	7339.0	4334.2	6.5848	30
23	Cal Poly San Luis Obispo	NaN	NaN	NaN	0
24	San Diego State University	24064.0	37041.7	7.8765	30
25	San Jose State University	2217.0	2222.9	2.8431	30
26	CSU Long Beach	NaN	NaN	NaN	0
27	CSU Fullerton	7533.0	42562.1	12.3274	30
28	Sacramento State	1054.0	NaN	NaN	0
29	San Francisco State University	NaN	NaN	NaN	0
30	Harvard University	854518.0	80220.1	7.3705	30
31	New York University	127660.0	7137.9	10.2187	30
32	University of Chicago	16357.0	50987.4	20.7787	30
33	MIT	102201.0	185444.2	9.5785	30
34	Johns Hopkins University	6204.0	19405.8	2.7755	28

TikTok Scale

Harvard: 854,518 followers
 Ohio State: 348,821
 Colorado: 273,678
 UCLA: 213,578
 Pepperdine: 7,943 (41st %ile)

Engagement Leaders

U Chicago: 20.78%
 UC Santa Cruz: 16.78%
 USF: 14.94% · U Miami: 14.0%
 UCSB: 12.14% · CSU Fullerton: 12.33%
 Pepperdine: 4.55% (21st %ile)

 Small accounts CAN win on engagement — content quality beats audience size

YouTube: Subscribers, Shorts Strategy & Posting Velocity

YouTube reveals content strategy — Shorts % and posting cadence are key differentiators

CA Schools — Social Media vs Admissions:

	INSTNM	ig_k	yt_k	tt_k	accept_rate	yield_rate	applications
	University of California-Los Angeles	509.0	410.0	213.6	9.0	50.4	146272
	University of California-Berkeley	416.0	0.1	32.5	11.0	46.0	124242
	Stanford University	270.0	77.5	208.7	3.6	81.9	57326
	San Jose State University	266.0	19.6	37.9	84.6	14.7	37132
	University of California-San Diego	153.0	7.1	23.3	26.7	20.4	134450
	University of California-Riverside	135.0	15.0	15.1	76.4	12.2	58048
	California State University-Sacramento	123.0	12.0	1.2	94.0	15.3	27702
	University of California-Santa Barbara	105.0	7.6	20.0	33.0	13.8	110259
	California State University-Fullerton	97.0	5.7	NaN	90.5	14.2	53559
	California State University-Bakersfield	89.0	0.5	NaN	93.9	10.4	13229
	University of California-Davis	76.0	1.4	7.5	41.8	16.4	98864
	University of California-Santa Cruz	75.0	0.8	2.2	65.8	9.3	71722
	Pepperdine University	64.0	17.0	7.9	62.9	11.6	11526
	Chapman University	55.0	7.6	6.0	65.4	15.9	15634
	University of La Verne	50.0	8.9	6.7	71.2	9.8	7110
	Biola University	35.0	155.0	2.1	74.4	29.7	3781
	Azusa Pacific University	23.0	2.1	4.1	88.4	13.7	3404
	Harvey Mudd College	21.0	5.9	2.2	12.7	35.8	5094
	University of Redlands	18.0	2.4	NaN	82.7	15.9	3766
	California State University-Los Angeles	13.0	2.2	3.2	91.3	9.4	32723
	California State Polytechnic University-Humboldt	12.0	4.7	NaN	98.2	5.9	15725

YouTube Snapshot

Pepperdine YouTube:

- 17,000 subscribers (61st %ile) ●
- 40.9% Shorts (83rd %ile) ●
- 69.9 views/day velocity (83rd %ile) ●
- 268 days avg between posts ● (91st percentile WORST)

Top YT Shorts users:

Biola: 100% · CSU Humboldt: 96.7%

CSU LA: 83.3% · Ohio State: 66.7%

Pepperdine: 40.9%

Critical Gap

Pepperdine posts on YouTube once every 268 days on average.

Dataset average: 58.4 days

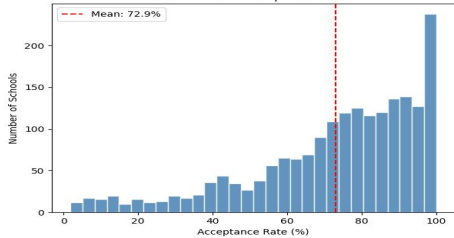
Pepperdine percentile: 91st WORST

YT view velocity predicts applications
 $r = +0.363$ ** (statistically significant)

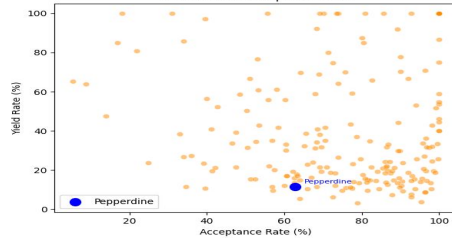
Admissions Context: Where Does Pepperdine Stand?

University Admissions & Social Media Analysis

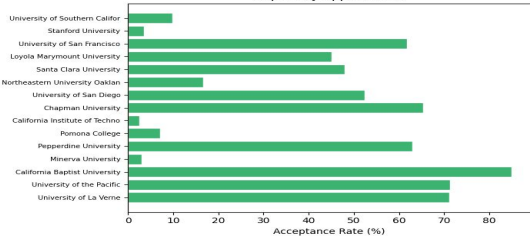
Distribution of Acceptance Rates



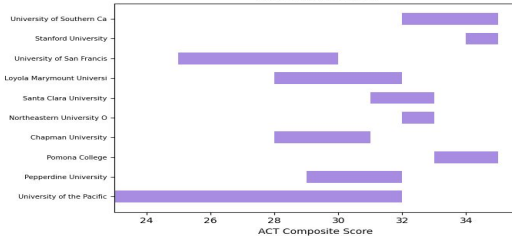
Yield Rate vs Acceptance Rate



CA Private Schools: Acceptance Rate (Top 15 by Applications)



ACT Score Range (25th-75th %ile) CA Private Schools



62.9%

Acceptance Rate
55th percentile

11.6%

Yield Rate
13th percentile ●

11,526

Applications
13th percentile ●

Low yield = admitted students choosing other schools. Social media is a potential lever.

CA Private Schools (SQL query result):									
		INSTNM	STABBR	CONTROL	APPLCN	acceptance_rate	yield_rate	ACTCM25	ACTCM75
0	University of Southern California	CA	2	82027	9.8	43.3	32.0	35.0	35.0
1	Stanford University	CA	2	57326	3.6	81.9	34.0	30.0	30.0
2	University of San Francisco	CA	2	24888	61.7	6.0	25.0	30.0	30.0
3	Loyola Marymount University	CA	2	23089	45.1	14.8	28.0	32.0	32.0
4	Santa Clara University	CA	2	18970	48.0	17.7	31.0	33.0	33.0
5	Northeastern University Oakland	CA	2	18030	16.7	14.9	32.0	33.0	33.0
6	University of San Diego	CA	2	17008	52.4	12.4	NAN	NAN	NAN
7	Chapman University	CA	2	15634	65.4	15.9	28.0	31.0	31.0
8	California Institute of Technology	CA	2	13856	2.6	61.2	NAN	NAN	NAN
9	Pomona College	CA	2	12249	7.1	50.2	33.0	35.0	35.0
10	Pepperdine University	CA	2	11526	62.9	11.6	29.0	32.0	32.0
11	Minerva University	CA	2	10864	3.0	43.6	NAN	NAN	NAN
12	California Baptist University	CA	2	7990	84.9	23.8	NAN	NAN	NAN
13	University of the Pacific	CA	2	7780	71.3	13.8	23.0	32.0	32.0
14	University of La Verne	CA	2	7110	71.2	9.8	18.0	25.0	25.0
15	Claremont McKenna College	CA	2	6529	9.6	53.8	33.0	35.0	35.0
16	Occidental College	CA	2	6461	44.2	17.8	31.0	34.0	34.0
17	California Lutheran University	CA	2	6005	75.7	11.0	24.0	31.0	31.0
18	Saint Mary's College of California	CA	2	5175	86.5	9.5	NAN	NAN	NAN
19	Harvey Mudd College	CA	2	5094	12.7	35.8	34.0	36.0	36.0
Top 10 Schools by Applications:									
		INSTNM	APPLCN	acceptance_rate	yield_rate				
0	University of California-Los Angeles		146272	9.0	50.4				
1	University of California-San Diego		134450	16.7	20.4				
2	University of California-Berkeley		124242	11.0	46.0				
3	University of California-Irvine		122699	28.6	19.2				
4	New York University		110807	9.2	51.4				
5	University of California-Santa Barbara		110259	33.0	13.8				
6	University of California-Davis		98864	41.8	16.4				
7	Northeastern University		98425	5.2	53.8				
8	University of Michigan-Ann Arbor		98310	15.6	47.3				
9	San Diego State University		89972	36.2	20.4				

Key Findings: What Correlates with Admissions Outcomes?

Pearson r correlations · ** $p < 0.05$ · * $p < 0.10$ · 38 universities

Predicts YIELD RATE

Higher social presence = more admitted students enroll

IG Followers → Yield Rate **+0.530**

**

TikTok Followers → Yield Rate **+0.481**

**

Total Social → Yield Rate **+0.424**

**

YT Total Views → Yield Rate **+0.334**

**

YT Subscribers → Yield Rate **+0.273**

*

IG Engagement → Yield Rate **-0.451**

** (negative — see slide 13)

Predicts APPLICATIONS

Content quality and virality drives application volume

TikTok Avg Likes → Apps **+0.500**

**

TikTok Avg Shares → Apps **+0.456**

**

IG Avg Comments → Apps **+0.402**

**

YT View Velocity → Apps **+0.363**

**

YT Shorts % → Apps **-0.293**

* (negative)

Predicts ACCEPTANCE RATE

More social presence = more selective (negative = good brand)

IG Followers → Acceptance **-0.390**

**

Total Social → Acceptance **-0.365**

**

YT Total Views → Acceptance **-0.318**

*

YT Subscribers → Acceptance **-0.276**

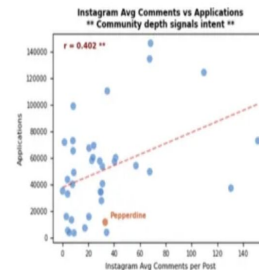
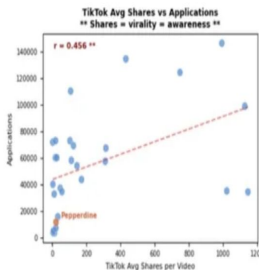
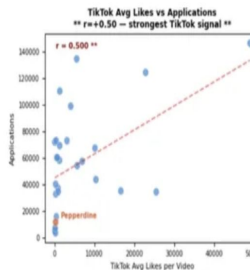
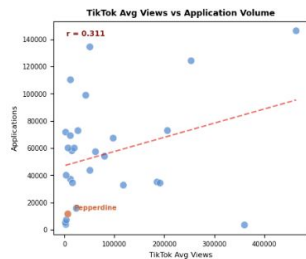
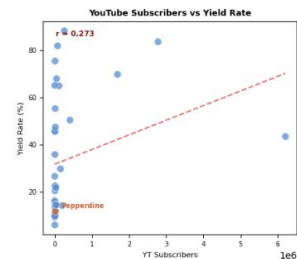
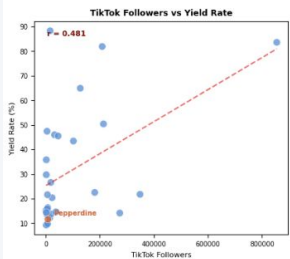
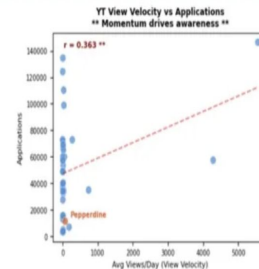
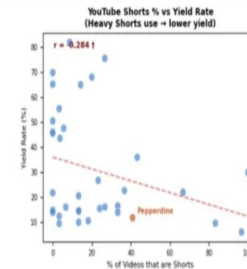
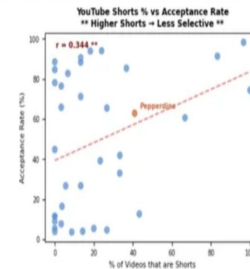
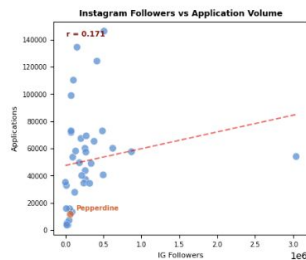
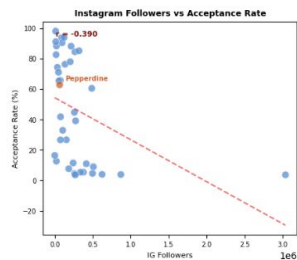
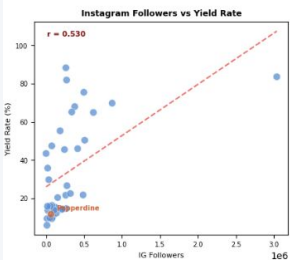
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YT Shorts % → Acceptance **+0.344**

** (positive)

The Visual Evidence: Scatter Plots from the Analysis

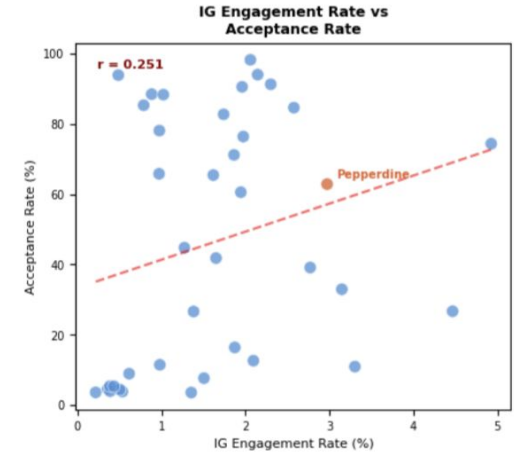
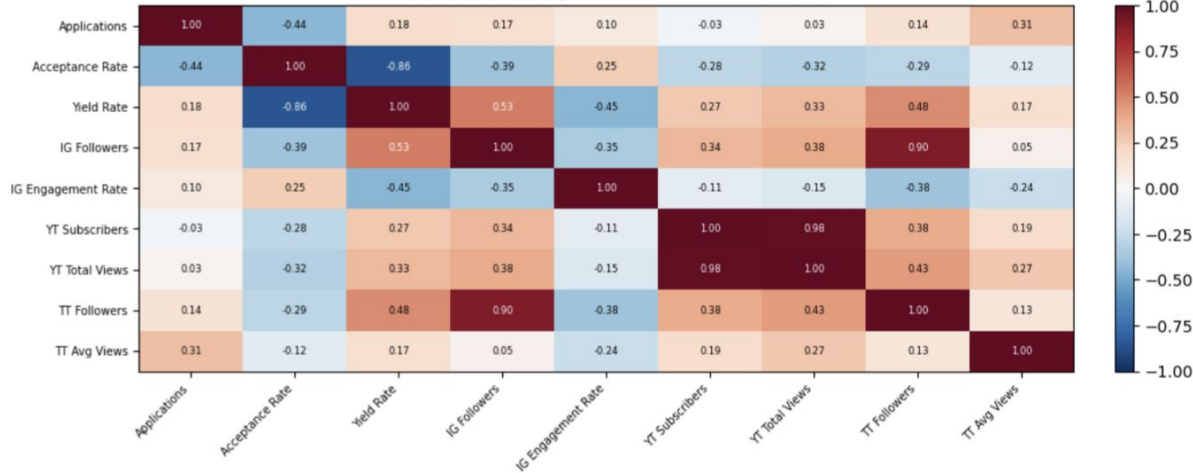
All charts produced by Cayden's notebook — Pepperdine labeled as orange dot in each



Correlation Heatmap: All Social Media Metrics vs. Admissions

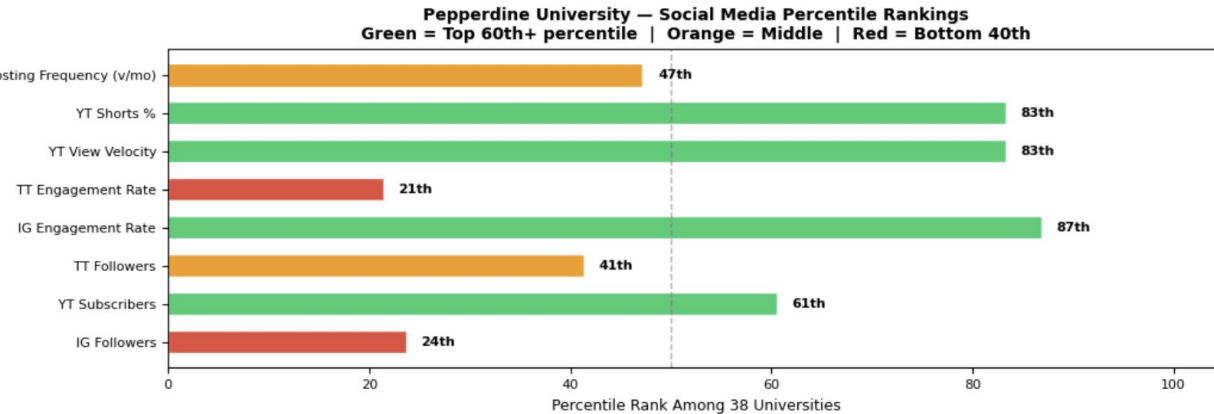
Dark red = strong positive · Dark blue = strong negative · White = no relationship

Correlation Heatmap: Social Media vs Admissions



Pepperdine Deep Dive: Percentile Rankings Among 38 Universities

Green = Top 60th+ · Orange = Middle · Red = Bottom 40th



✓ Strengths

- IG Engagement Rate: 87th %ile
- YT Shorts %: 83rd %ile
- YT View Velocity: 83rd %ile
- YT Subscribers: 61st %ile

● Gaps to Close

- IG Followers: 24th %ile
- Yield Rate: 13th %ile
- YT Posting: 91st WORST
- TikTok Eng. Rate: 21st %ile

Pepperdine: Full Social Media Stats at a Glance

All numbers from the scraped dataset compared to the 38-school dataset average

64,828

Instagram Followers
Avg: 286,709 · 24th %ile

2.97%

IG Engagement Rate
Avg: 1.6% · 87th %ile 

1,894

IG Avg Likes/Post
avg likes per post

7,943

TikTok Followers
Avg: 88,842 · 41st %ile

4.55%

TikTok Engagement Rate
Avg: 8.2% · 21st %ile

6,741

TikTok Avg Views
avg views per video

17,000

YouTube Subscribers
Avg: 321,202 · 61st %ile 

40.9%

YouTube Shorts %
Avg: 21.5% · 83rd %ile 

268 days

Avg Days Between YT Posts
Avg: 58.4 days · 91st worst

62.9%

Acceptance Rate
55th percentile

11.6%

Yield Rate
Avg: 33.7% · 13th %ile 

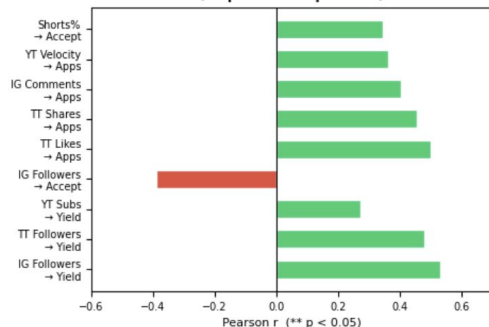
11,526

Application Volume
Avg: 50,971 · 13th %ile 

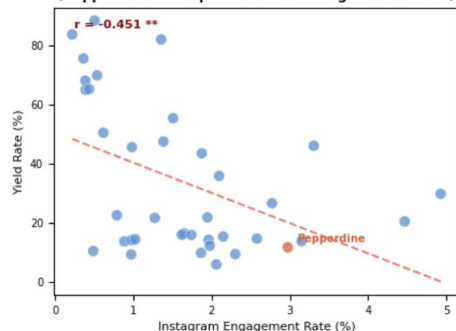
Interesting Findings & Nuances

Counterintuitive patterns the data revealed

Key Significant Correlations
(all $p < 0.05$ or $p < 0.10$)



IG Engagement Rate vs Yield Rate
(Pepperdine: 87th percentile — strength to build on)



What These Anomalies Mean

High Engagement ≠ High Yield

IG engagement rate is negatively correlated with yield ($r = -0.451^{**}$). Highly engaged audiences tend to be niche. Scale and reach matter more for yield.

YouTube Shorts → Less Selective

More Shorts % positively correlates with acceptance rate ($r = +0.344^{**}$). Elite schools post long-form — they don't need viral hooks to attract applicants.

TikTok Likes Beat Followers

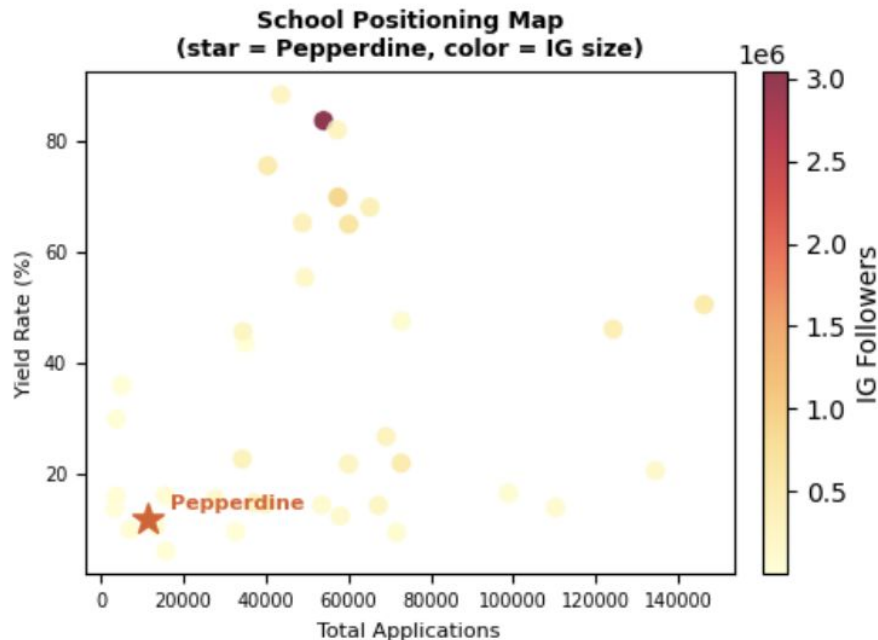
TikTok avg likes ($r = +0.500$) predicts applications far better than follower count ($r = +0.141$, not significant). Content depth beats audience size.

Posting Frequency Barely Matters

Posting frequency vs yield rate: $r = +0.114$ (not significant). It's not about how often — it's about quality and momentum (view velocity).

Pepperdine vs. Peers: Positioning Map

Where Pepperdine sits relative to all 38 schools on yield rate and application volume



What This Map Shows

Top-right schools

High yield + high applications
→ Harvard, Stanford, Yale
→ Strong social AND strong brand

Bottom-left = Pepperdine

Low yield (11.6%) AND low applications (11,526)
→ Admitted students choosing elsewhere
→ Social reach is part of the solution

The goal

Move Pepperdine up and right
on this chart over 3-5 years
by growing social reach
and improving content strategy

Dot color = IG size

Larger/darker dots = more
Instagram followers — the schools
with most IG reach tend to cluster
in the top-right quadrant

Strategic Recommendations for Pepperdine

Based directly on the percentile gaps and correlation findings



1. Grow Instagram Following

HIGH PRIORITY

At 24th percentile, IG followers is our biggest gap. IG followers → yield ($r=+0.530^{**}$). This is the #1 lever.



2. Fix YouTube Posting Cadence

HIGH PRIORITY

268 days between posts (91st worst) wastes our strong Shorts % and view velocity. Target 2-4 posts/month.



3. Prioritize TikTok Content Quality

MEDIUM

TikTok avg likes → applications ($r=+0.500^{**}$). Focus on content that earns likes and shares, not just follower campaigns.



4. Leverage Student Creators

QUICK WIN

Our 87th %ile IG engagement shows the audience is receptive. Amplify authentic student voices — it works.



5. Cross-Platform Repurposing

QUICK WIN

Top YT Shorts → TikTok + IG Reels. The content already exists. Systematize distribution across all platforms.



6. Tie Social KPIs to Enrollment Goals

STRATEGIC

Build a quarterly dashboard tracking these metrics alongside admissions data. Make social media a business metric.

Limitations & Future Research

Being honest about what the data can and cannot tell us

Current Limitations

- **Recent Posts Only:**

TikTok ~30, Instagram ~12 posts — reflects now, not historical trends or full 2024

- **Correlation $\neq$ Causation:**

We show associations — not proof social media drives admissions. Prestigious schools may just have both

- **38 of 51 Merged:**

13 schools incomplete — may bias toward larger, more active institutions

- **API Rate Limits:**

Multiple RapidAPI accounts needed; some data may be slightly stale

Future Research

- **Longitudinal Tracking:**

Scrape same schools quarterly — does social growth actually precede application changes?

- **Content Analysis via AI:**

Use NLP to analyze what content types (campus life vs. academics) drive most engagement

- **Admitted Student Survey:**

Ask Pepperdine's incoming class: did social media influence their decision to apply/enroll?

- **Peer Competitor Deep Dive:**

Compare Pepperdine directly vs LMU, USD, Chapman, Santa Clara — direct CA private peers

- **YouTube Watch Time:**

Track completion rates, not just views — deeper engagement may be stronger admissions signal

Thank You

Key Takeaways

Instagram followers most strongly predict yield rate ($r=+0.530^{**}$) — Pepperdine is at only the 24th percentile

TikTok content quality (avg likes & shares) predicts application volume better than follower count

YouTube view velocity and posting consistency are underutilized — Pepperdine posts every 268 days on average

Pepperdine's engagement quality is strong (87th %ile IG) but reach is weak — the strategy works, it needs scale

Social media is measurably connected to admissions outcomes — these are business metrics, not vanity metrics